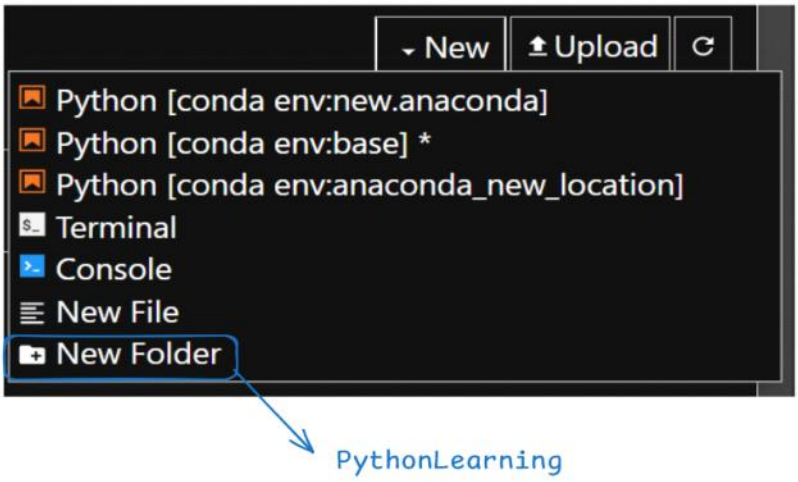
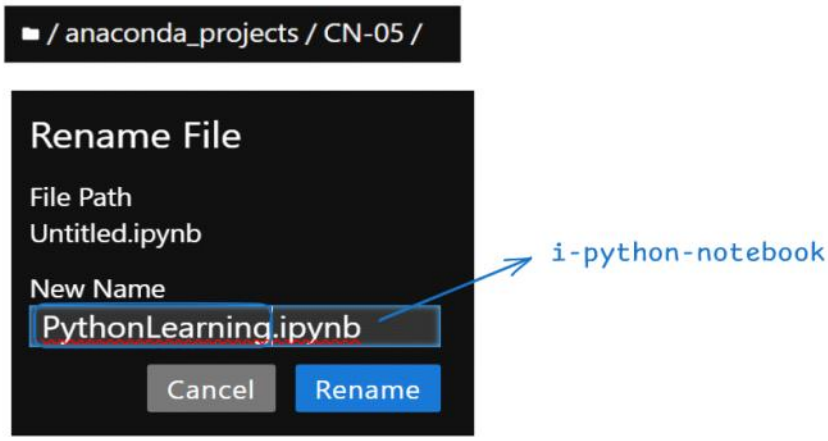
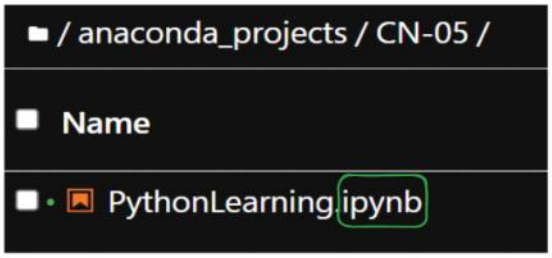
Introduction to Python-II

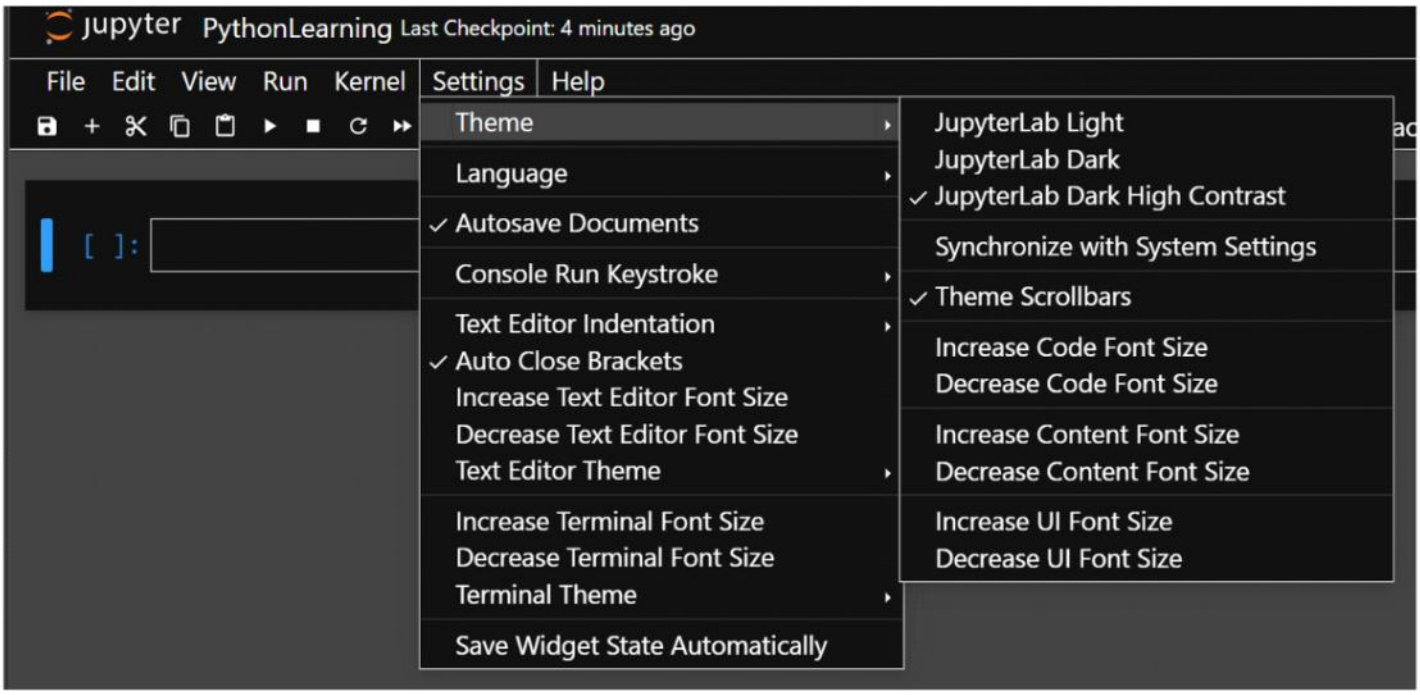
Session Objectives

- Discover Python's features
- Understand how Python differs from Excel and SQL
- Learn Python's role in Data Science
- Understand the basic syntax of Python
- Learn about variables and their usage
- Declare and assign values to variables
- Explore data types, check them, and perform type conversion








jupyter PythonLearning Last Checkpoint: 4 minutes ago
File Edit View Run Kernel Settings Help
[ ]:
Theme
Language
Autosave Documents
Console Run Keystroke
Text Editor Indentation
Auto Close Brackets
Increase Text Editor Font Size
Decrease Text Editor Font Size
Text Editor Theme
Increase Terminal Font Size
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Terminal Theme
Save Widget State Automatically
JupyterLab Light
JupyterLab Dark
JupyterLab Dark High Contrast
Synchronize with System Settings
Theme Scrollbars
Increase Code Font Size
Decrease Code Font Size
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Code
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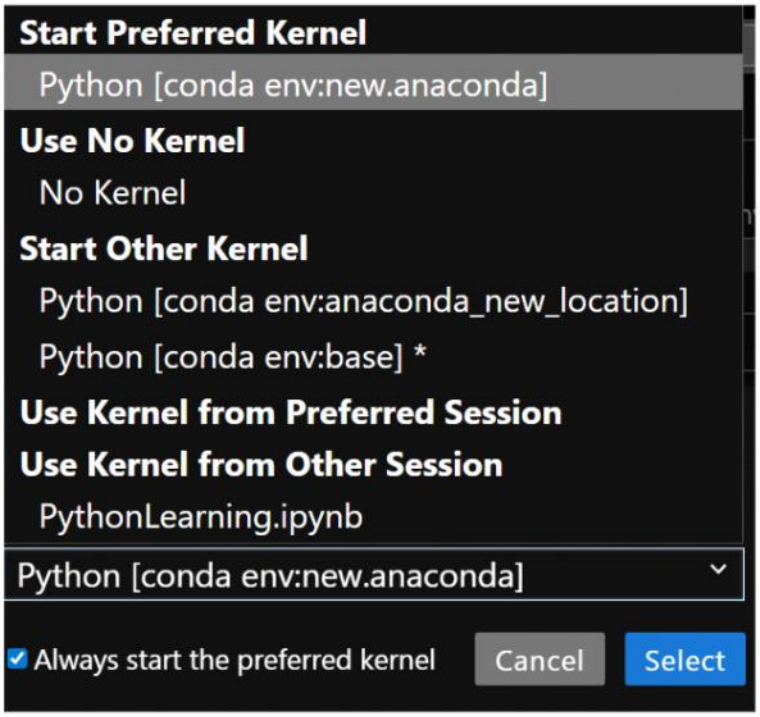
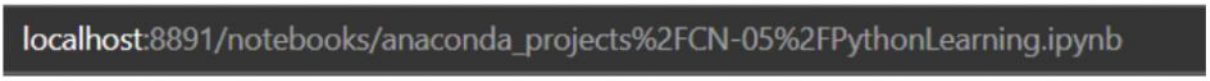
Code
Markdown
Raw
Python Code -> Interpreter
readme.md
Raw content


Start Preferred Kernel
Python [conda env:new.anaconda]
Use No Kernel
No Kernel
Start Other Kernel
Python [conda env:anaconda_new_location]
Python [conda env:base] *
Use Kernel from Preferred Session
Use Kernel from Other Session
PythonLearning.ipynb
Python [conda env:new.anaconda]
Always start the preferred kernel
Cancel
Select

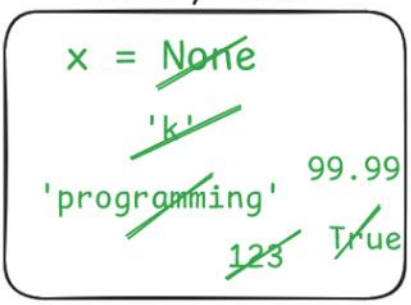

localhost:8891/notebooks/anaconda_projects%2FCN-05%2FPythonLearning.ipynb

Behind the Scene

TOP BOTTOM

```
x = 'k' # char
x = 'Programming' # text
x = 123 # Numeric
x = True # Boolean
x = 99.99 # Float
print(x)
print(type(x)) # Float
```

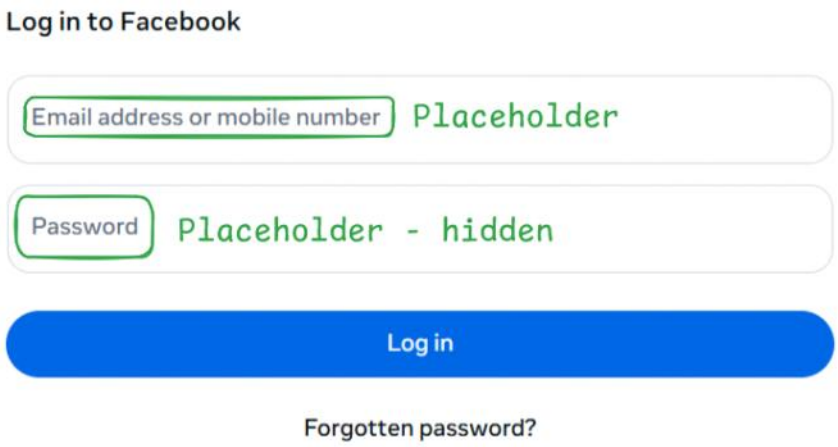
Memory View



Console

```
99.99
float
```

Input()



input function is to take input from user and store in a variable

```
user_name = input("Enter the UserName: ")
print(user_name)
```

```
Enter the UserName:  UltimateForce
UltimateForce
```

Behind the Scene

Console

dynamic input

```
Enter the UserName : ______
UltimateForce
```

Memory

```
user_name = None
            UltimateForce
```

```python
print("Hello World")
```

```
Hello World
```

Python Programming

Let's Learn Together & Grow Together!!

1. Learn Faster!
2. Grow Faster!

- Its a Programming Language
  - It's a High Level Language
    - Easy To Understand. (Human Readable!)
- Handles Large Dataset
- It's an interpreter Language
- Rich in Libraries
- Amazing Support from the Community...
- Its an platform Independent -> .byte code

```python
# This is a Single Line Comment
x = 'k' # char
x = 'Programming' # text
x = 123 # Numeric
x = True # Boolean
x = 99.99 # Float
print(x)
print(type(x)) # Float
```

```
99.99
<class 'float'>
```

```
# Java Language
int x = 99
System.out.println(x) # 99
```

```python
x = 9+6j
print(x)
print(type(x))
```

```
(9+6j)
<class 'complex'>
```

```python
x = int('10') + 1
print(x) # 11
print(type(x)) # int
```

```
11
<class 'int'>
```

```python
# Parameters '\n' by default
print("Python")
print("Programming")
```

```
Python
Programming
```

```python
print("Python" , end = " ")
print("Programming") # end = '\n'
print("Coding is Awesome 😎")
```

```
Python Programming
Coding is Awesome 😎
```

```python
print("Python" , end = "\t")
print("Programming") # end = '\n'
print("Coding is Awesome 😎")
```

```
Python	Programming
Coding is Awesome 😎
```

```python
# Separator -> Delimiter [.csv] [',']
print("Mon","Tues","Wed","Thurs","Fri")
```

```
Mon Tues Wed Thurs Fri
```

```python
print("Mon","Tues","Wed","Thurs","Fri" , sep = " - ")
```

```
Mon - Tues - Wed - Thurs - Fri
```

```python
print("Mon","Tues","Wed","Thurs","Fri" , sep = " -> ")
```

```
Mon -> Tues -> Wed -> Thurs -> Fri
```

```
Mon -> Tues -> Wed -> Thurs -> Fri
```

Input()

- Used to accept user input from the *keyboard* as a **string**, which can be further type cast as per requirement..
- `user_name = input("Enter the Prompt....")`

```
user_name = input("Enter the UserName: ")
print(user_name)
```

```
Enter the UserName:  UltimateForce
UltimateForce
```

```
user_name = input("Enter the UserName: ")
print(user_name)
```

```
Enter the UserName:  Maverick
Maverick
```

```
# Type Cast
val = input("Enter the Value: ") # int('11') -> 11
print(val)
print(type(val))
```

```
Enter the Value:  11
11
<class 'str'>
```

```
val = int(input("Enter the Value: ")) # int('11') -> 11
print(val)
print(type(val))
```

```
Enter the Value:  11
11
<class 'int'>
```

```
val = int(input("Enter the Value: ")) # int('abc') -> Error
print(val)
print(type(val))
ValueError: invalid literal for int() with base 10: 'abc'
```

```
print(val * 11) # val = 11 * 11 = 121
```

```
121
```

```
print(val) # 11
```

```
11
```

```
val = val * 11
# val *= 11
print(val)
```

```
121
```

```
x = int(input("Enter the value between 1 to 100 :"))
x = x + 10
print(x)
```

```
Enter the value between 1 to 100 : 67
77
```

```
x = input("Enter the value between 1 to 100 :")
x = x + 10
print(x)
print(type(x))
TypeError: can only concatenate str (not "int") to str
```

```
x = int(input("Enter the value between 1 to 100 :"))
if x > 100:
    print("Invalid Input, Please Enter the Val Between 1-100")
else:
    x = x + 10
    print(x)
```

```
Enter the value between 1 to 100 : 111
Invalid Input, Please Enter the Val Between 1-100
```

```
x = int(input("Enter the value between 1 to 100 :"))
if x > 100:
    print("Invalid Input, Please Enter the Val Between 1-100")
else:
    x = x + 10
    print(x)
```

```
Enter the value between 1 to 100 : 91
101
```

```
# Java Language
if x>100{
    print("Invalid Input, Please Enter the Val Between 1-100")
}
```

```python
# String Pattern
user_name = input("Enter the UserName: ")
email_id = input("Enter the Email Address: ")
print("Hey " + user_name + " Welcome to the Python Course!, You are being registered with us " + email_id + " .
```

```
Enter the UserName:  Siddharth
Enter the Email Address:  siddharth1@gmail.com
Hey Siddharth Welcome to the Python Course!, You are being registered with us siddharth1@gmail.com .
```

```python
# String Pattern f-string
user_name = input("Enter the UserName: ")
email_id = input("Enter the Email Address: ")
print(f"Hey {user_name}, Welcome to the Python Course!, You are being registered with us {email_id} ✨" )
```

```
Enter the UserName:  Siddharth
Enter the Email Address:  siddharth1@gmail.com
Hey Siddharth, Welcome to the Python Course!, You are being registered with us siddharth1@gmail.com ✨
```

```python
# This is a Single Line Comment
print("Hello World!")
```

```
Hello World!
```

```python
# Multiline Comment
"""
    This is a Multiline Comment.....
    Put Some Important Stuff here relevant
    To Your Python Code.
"""
num1 = int(input("Enter the First Value: "))
num2 = int(input("Enter the Second Value: "))
result = num1 + num2
print(f"The final Result is {result}")
```

```
Enter the First Value:  77
Enter the Second Value:  22
The final Result is 99
```

```python
print("Hey: What's your name: \t Jhanvi Mundra")
print("Hello\nWorld")
print("x" , end = " ")
print("y" , end = " ")
print("z" , end = "\n")
print("Welcome Back!")
```

```
Hey: What's your name:   Jhanvi Mundra
Hello
World
x y z
Welcome Back!
```